Causal Information Outside a Frozen Board Probe in Othello-GPT

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Abstract

Linear probes recover Othello-GPT's board with high accuracy, and editing along probe directions changes the model's legal-move predictions. Neither fact shows that a frozen probe captures every residual-stream distinction the network uses. We study one 8-layer Othello-GPT, one frozen 64-square linear board probe, and residual edits at `blocks.6.hook_resid_mid` that lie in the kernel of the probe's class-logit contrasts. The matched-L2, matched-divergence, and donor-free edits start from the Editor A reference activation, which is the natural recipient activation after a full-space board editor, with its manifold penalty off, has been applied. The switching study adds its edit to the natural recipient instead. Kernel steps of similar length barely move the output at a natural donor activation. The edit leaves the probe's 64×3 probability tensor at that reference unchanged to within 2.2×10^{-14}. On 48 recipient–target pairs, kernel directions taken from the intended target's natural activation raised a legal-mass response score more than wrong-target and random kernel directions. This held at matched residual L2 and again when every direction was scaled to the same output Jensen–Shannon divergence (0.01) from the Editor A reference. At matched divergence the advantage over wrong targets was small (+0.0266, 95% interval $[0.0179, 0.0357]$). A 64-component ridge map from symbolic board features, trained only on training games, produced an advantage on 48 new pairs without using the target's activation at evaluation (+0.0201 versus wrong targets). Mean scores nevertheless stayed negative: the learned edit left more probability on moves that should have become illegal than on moves that should have become legal in 47 of 48 pairs. About 56% of the map's mean gain over Editor A was already obtained by a map that sees only player and occupancy. That ratio is not a stage-specific correction. On a fresh cohort, both target-switching contrasts were positive on average, but both were positive together in only 13 of 32 triplets, below a preregistered 60% gate. Those two cohorts are the only fresh samples. The positive results reuse validation pairs. A factorial test of visible–kernel interaction gave estimates near zero. The evidence supports a narrow conclusion: this frozen probe does not exhaust the residual information that affects legal-move outputs, part of the remainder is predictable from symbolic state, and the recovered map is not a transferable board state. The matched-L2, matched-divergence, and donor-free results are recomputed from saved arrays in the accompanying package. The ablation, switching, and factorial results are archival values whose arrays were not available for recomputation.

1 Introduction

A linear probe that reads the board from Othello-GPT is easy to treat as a description of what the model knows. Li et al. (2023) trained a GPT-style network on legal Othello move sequences, showed that nonlinear probes recover the board, and showed that editing the probed representation changes which moves the network treats as legal. Nanda, Lee, and Wattenberg (2023) then showed that a player-relative linear probe is enough to read the board, and that adding probe directions is

enough to steer play. Those results give a causal role to a decoded board. They do not show that one frozen probe is a complete causal summary of the residual stream.

Probing and intervention answer different questions. A probe asks whether a feature is linearly readable. An intervention asks whether a specified change to an activation changes the frozen network's output. Elazar et al. (2021) made the gap explicit by removing linearly readable information and measuring behavior; high probe accuracy did not imply that the probed feature was what the model used. The complementary experiment adds information that the probe cannot see, holds the probe's readout fixed, and asks whether behavior still moves.

We run that experiment on the Nanda redistribution of Othello-GPT, an 8-layer Transformer with residual width 512. We freeze a 64-square multinomial linear probe at the last real token of `blocks.6.hook_resid_mid`. The probe's independent class-logit contrasts define a rowspace and a complementary nullspace. Adding a nullspace vector to an activation leaves every square's probe probabilities at that activation unchanged, up to floating-point error. We call such an edit probe-preserving relative to this probe. The phrase does not mean that no decoder could read the change, and it does not mean that the model has a second hidden board.

Every edit in this paper is applied to a specific reference activation, and the choice matters. For a recipient position we write h_R for its natural activation. Editor A, a full-space residual editor from earlier work on this model, moves h_R toward the intended target board as read by the probe; we call its output h_E the Editor A reference activation. Its manifold penalty was off. A natural activation from a position that actually has the target board is h_D. In the matched-L2, matched-divergence, and donor-free studies the final patched activation is $h_E + t\,\delta$ with δ in the probe nullspace. The switching study adds its edit to h_R. Probe preservation and the output-divergence calibration are both measured against h_E, not against h_R. The experiments therefore ask a specific question. Start from an activation whose probe readout has already been pushed toward the target board. Can directions that this probe cannot see shift legal-move probability further toward the target, and do they do so more for the intended target than for other boards?

The output measure is a difference of probability masses. For a recipient position with legal set L_r and an intended target with legal set L_t, G is the model's probability on moves that became legal, Q is its probability on moves that became illegal, and $Y = G - Q$. A higher Y means probability moved toward the target's legal set and away from the recipient's. It is not move accuracy, and a higher Y is not a claim that the model now plays the target board.

We ask three questions. First, do nullspace directions taken from the target's natural activation change Y more than wrong-target and random nullspace directions? We answer under two different strength controls, matched residual L2 and matched output Jensen–Shannon (JS) divergence. Second, can a map that sees only symbolic features of the target produce a comparable nullspace direction without the target's activation at evaluation? Those features are player-relative occupancy, the legal-move mask, the player to move, and the game stage. Third, how far does that map go? We report how much of its gain is already present in a stage-only map. We also report two tests that a transferable board state would be expected to pass and that this map failed: reliable switching between two targets on the same recipient, and an interaction between independently set visible and nullspace components.

The contribution is empirical and narrow. We do not propose a theory of Transformer decisions. We do not claim that every board probe would miss this information, or that the 64-component map recovered the model's decision state. The experiments support three statements about this probe and this model. Directions that the probe cannot distinguish still carry some target-aligned influence on legal-move predictions. Part of that influence can be predicted from symbolic state

without a target activation at evaluation. The resulting edit is far from complete state transfer.

2 Related work

Probing treats a decoder's success as evidence that a representation contains a variable, and the methodological literature has long warned that readout is not use. Hewitt and Liang (2019) introduced control tasks that a probe can solve without tracking linguistic structure. Belinkov (2022) surveyed the gap between encoding and causal role. Elazar et al. (2021) used iterative nullspace projection (Ravfogel et al., 2020) to remove a property and then asked whether behavior changed. Belrose et al. (2023) gave a closed-form linear erasure method with a guarantee against all linear classifiers. Ravfogel, Prasad, Linzen, and Goldberg (2021) used the same rowspace–nullspace split to build counterfactuals: their AlterRep method changes the rowspace component of a probe-identified feature, keeps the nullspace component, and measures the change in word predictions. Those papers remove or alter what a probe reads. Our edits do the reverse: they keep what the probe reads and change the rest. The papers above do not ask whether the orthogonal complement of a particular frozen probe still carries target-specific causal influence.

Othello-GPT is the setting in which world-model claims have been examined most carefully. Li et al. (2023) argued that a network trained only on move tokens maintains an internal board, using nonlinear probes and interventional edits. Nanda et al. (2023) showed that a player-relative linear basis makes the board linearly readable and steerable by vector addition. Hazineh, Zhang, and Chiu (2023) intervened along linear board representations layer by layer and found causal use concentrated in the middle layers. Those interventions move the model along the probe. Moving along the probe cannot tell us whether the probe is a sufficient statistic for legal-move behavior; that question requires moving within its kernel.

Two later Othello studies make the readout-versus-use distinction in other designs. Jacopin (2026) reproduced the linear player-relative board in a masked-diffusion language model trained on the same data. The representation transferred but was weaker, and a Li/Nanda-style board intervention had only a small, inconsistent effect relative to an autoregressive control. That study asks whether a readable world model is used under a different training objective. It does not freeze a probe and edit in that probe's kernel. Lin, JackS, Karvonen, and Can (2024), in a research note that has not been peer reviewed, report that Othello-GPT appears to compute legality through many localized heuristics, including neurons that respond to game stage. That account is compatible with our finding that a stage-only map carries much of the mean gain, but it is not a nullspace test of a frozen probe.

Activation steering in language models adds residual directions derived from contrastive prompts or trained readers (Turner et al., 2023; Zou et al., 2023; Rimsky et al., 2024), usually with a strength coefficient. We instead match the output divergence of several directions, including wrong-target directions, so that a direction cannot look better merely by disrupting the next-move distribution more. That is a matching condition, not a new steering algorithm. Makelov, Lange, Geiger, and Nanda (2024) show that a subspace intervention can activate pathways that are dormant on natural inputs, which is the relevant caution for edits at h_E.

Causal abstraction and distributed alignment search (Geiger et al., 2024, 2025) ask whether a high-level causal variable aligns with a distributed subspace under interchange interventions. We do not fit such an alignment and make no faithfulness claim. Our estimand is the change in Y under an explicitly constructed residual edit. It is a causal effect of that intervention, not evidence that we found the variable the network uses.

The closest prior statement is that linear board features in Othello-GPT are readable and steerable, while probes in general are not causal summaries. The design here differs from AlterRep in direction: AlterRep changes the rowspace and keeps the nullspace, and these experiments keep the probe readout and change the nullspace, with wrong-target and random controls and with the completeness tests reported whether they pass or fail. That description is not a claim that no earlier paper did something similar.

3 Experimental setup

3.1 Model, site, and probe

The model is the TransformerLens conversion of Nanda's 8-layer Othello-GPT ($d_{\text{model}} = 512$, 8 heads, vocabulary 61, context 59) trained on synthetic legal self-play following Li et al. (2023). Token 0 is pass and tokens 1–60 index the 60 playable squares in row-major order, skipping the four center squares. The model is never fine-tuned. The intervention site is the residual stream at `blocks.6.hook_resid_mid`, last real token. That site was fixed before the studies reported here; we did not search over layers.

The probe is a frozen set of 64 square-wise multinomial logistic regressions in the player-relative basis {empty, mine, theirs}, trained on training-split activations. For each square we form independent class-logit contrasts against a reference class and stack them into a contrast matrix $W \in \mathbb{R}^{124\times512}$. From the saved singular values, 124 exceed 10^{-8} times the largest, so the rowspace has dimension 124 and the nullspace has dimension 388. The saved nullspace projector P_N has trace 388, is idempotent to 10^{-15}, and has eigenvalues within 10^{-14} of 0 or 1. The projector was built so that $WP_N = 0$. The matrix W is not in this package, so that product is not recomputed here. What is checked is that the saved probe probabilities do not move. Adding $P_N\delta$ to an activation leaves every probe logit at that activation unchanged, and hence every probe probability. Figure 1 shows the geometry. The diagram describes the matched-L2, matched-divergence, and donor-free edits. The switching study applies its edit at h_R instead.

3.2 Activations and interventions

All studies use the same three activations for a recipient–target pair. The natural recipient activation h_R is the model's activation at the recipient position. The Editor A reference activation h_E is the output of Editor A applied to h_R with the target's player-relative board as its objective. Editor A is a full-space residual editor with L2 budget 16, 40 gradient steps, learning rate 0.4, and norm penalty 0.01; its own preregistered validation was negative. The natural donor activation h_D is the model's activation at the target position itself.

The correct-target nullspace direction is $\delta_{\text{correct}} = P_N(h_D - h_E)$. A wrong-target direction is $P_N(h_W - h_E)$ for a wrong-target donor h_W with the same player and similar occupancy, legal-set size, and prefix length. A random direction is P_N applied to a seeded Gaussian vector. In SPEC, the matched-divergence study, and NST, the final patched activation is $h_E + t\,\delta$. The switching study adds its nullspace step to h_R instead. The probe's readout at $h_E + t\,\delta$ equals its readout at h_E. It does not equal the readout at h_R: in the matched-divergence study the probe tensors at $h_E + t\,\delta$ and h_R differ by at least 0.93 in some entry for every intervention, because Editor A has already changed the decoded board.

Two strength controls are used. In the matched-L2 study (SPEC), wrong and random directions are

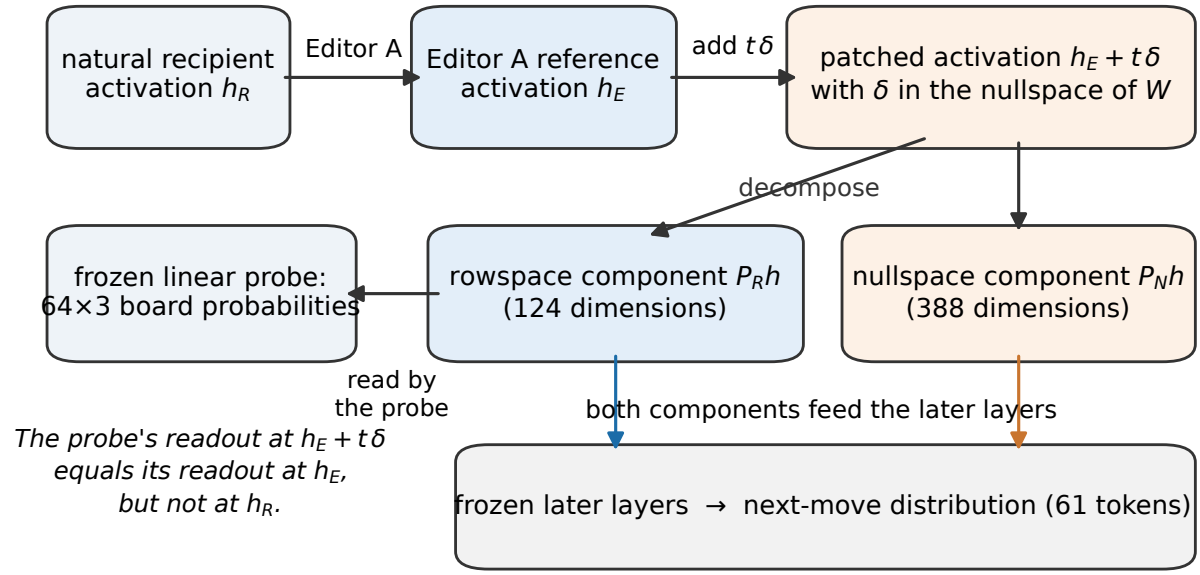


Figure 1. Activations and the probe geometry for the matched-L2, matched-divergence, and donor-free studies. The natural recipient activation h_R is first moved by Editor A to the Editor A reference activation h_E. Each of those edits adds a nullspace direction $t\,\delta$ to h_E. The switching study instead adds its edit to h_R. The frozen probe reads only the rowspace of its contrast matrix, so its readout at $h_E + t\,\delta$ equals its readout at h_E. Both rowspace and nullspace components feed the later layers of the frozen model, which produce the next-move distribution. The diagram does not claim that rowspace features are causally unused.

rescaled to the L2 norm of δ_{correct} and applied with $t = 1$. In the matched-divergence study, each direction is scaled independently so that $\text{JS}(F(h_E), F(h_E + t\,\delta)) = 0.01 \pm 0.0002$, where F maps an activation at the site to the model's next-move distribution. The search uses a frozen geometric grid from 0.003125 to 8 followed by 16 bisection steps and does not look at Y. The reference for this calibration is the Editor A reference activation. Measured from the natural recipient h_R, the same patched activations have mean JS 0.043.

The donor-free map (NST) is fitted on training-game activations only. Nullspace residuals $z = P_N h$ are modeled as $\mu + Va$ with $k = 64$ principal components, and a ridge regression ($\alpha = 10$) predicts the coefficients a from a 258-dimensional symbolic vector b. The vector holds 64×3 player-relative occupancy one-hots, a 64-entry legal-move mask, a player indicator, and occupancy divided by 64. Grouped cross-validation on training games chose k and α by residual R^2 before any held-out response was seen. The selected k is the largest value on that grid, and the cross-validated R^2 was still higher there than at the next smaller value. At evaluation the learned direction is $\delta_{\text{learned}} = P_N \hat{z}(b_t) - P_N h_E$, which uses the target's symbolic features and the Editor A reference activation but not h_D. The comparison directions are a wrong-target map direction $P_N \hat{z}(b_w) - P_N h_E$, a stage-mean direction, a random direction, and the oracle $P_N(h_D - h_E)$. The oracle was computed only after the learned directions were frozen. Every direction is calibrated to JS 0.01 from h_E as above.

3.3 Response score

For recipient legal set L_r and intended-target legal set L_t,

$$G = \sum_{s \in L_t \setminus L_r} p(s), \qquad Q = \sum_{s \in L_r \setminus L_t} p(s), \qquad Y = G - Q,$$

where p is the model's softmax over the 61-token vocabulary and s ranges over move tokens. Legal sets come from an Othello simulator. Every contrast compares two interventions on the same pair and scores both on the intended target's legal-set difference. A wrong-target intervention is therefore scored on the intended target's sets, not on the wrong board's own sets. JS divergence uses natural logarithms after adding 10^{-12} to each probability and renormalizing; as in the original code, each Kullback–Leibler term applies the same smoothing to its two arguments again, which changes JS by less than 10^{-9}. Total variation (TV) distance is half the L1 distance between two next-move distributions.

3.4 Pairs and triplets

Pairs require the same player to move, nonempty became-legal and became-illegal sets, matched occupancy, and disjoint recipient and target games. Category A pairs have board Hamming distance greater than 8 and category B pairs have distance from 2 to 8. The short names below are FIBER for the kernel diagnostic, SPEC for the matched-L2 study, NST for the donor-free map, SEM for the feature ablation, FRESH for target switching, and VNF for the factorial test. SEM is not a standard error. SPEC froze 32 A and 16 B pairs from unused validation games (pair-manifest SHA-256 `2789accf…1143d3`). The matched-divergence study reuses exactly those 48 pairs, so it is a follow-up on the same sample, not a new one. NST froze 48 different validation pairs (hash `39abdc79…5092e6`) after excluding every game used by earlier studies. A symmetric-target design on the remaining validation games was infeasible because no category-B triplets existed. The switching study (FRESH) and the factorial study (VNF) therefore each generated a new 400-game simulator cohort (seeds 9182 and 10182) and selected 16 A and 16 B triplets with 96 distinct game roles. The sealed confirmation split was never opened.

3.5 Statistics

The statistical unit is the pair or triplet. The matched-divergence study contains 336 calibrated interventions, seven per pair, but the three wrong-target and three random responses are averaged within each pair before any interval is computed, so its sample size is 48. Intervals are 95% percentile bootstrap intervals clustered on recipient game, with 2000 replicates from seed 82 for SPEC, the matched-divergence study, and NST. FRESH, VNF, and the feature-ablation study used 5000 replicates. In SPEC and NST each recipient game appears in exactly one pair, so clustering on recipient game amounts to resampling pairs. Target games are not clustered. In SPEC, 31 of the 48 pairs share a wrong-donor game with another pair, and the largest linked group has 7 pairs. The bootstrap does not model that dependence. The SPEC, NST, FRESH, and VNF floors were written into preregistrations before those studies' outcomes. The fixed-divergence floor of 0.02 was not: the first frozen matched-divergence protocol matched the wrong and random directions to the correct direction's own divergence from h_E and required a mean contrast of at least 0.05, and the fixed-0.01 protocol was committed after that run had been recorded as inconclusive. Subgroup intervals are descriptive. No multiplicity adjustment is applied. We report means, intervals, and win counts on the Y scale and do not call an effect large because its interval excludes zero.

Table 1. Headline contrasts and their evidence level. Means are on the Y scale. Intervals are 95% percentile bootstrap intervals clustered on recipient game. Recomputed values were derived again from saved arrays; archival values are verbatim console output from the original runs.

Study	Unit	Contrast	Mean	95% interval	Evidence
SPEC (matched L2)	48 pairs	correct − wrong	+0.3916	[0.3175, 0.4758]	Recomputed
Matched JS	48 pairs	correct − wrong	+0.0266	[0.0179, 0.0357]	Recomputed
Matched JS	48 pairs	correct − random	+0.0230	[0.0103, 0.0360]	Recomputed
NST (donor-free)	48 pairs	learned − wrong	+0.0201	[0.0108, 0.0298]	Recomputed
NST (donor-free)	48 pairs	learned − Editor A	+0.0537	[0.0452, 0.0618]	Recomputed
SEM ablation	48 pairs	FULL − LEGAL_ONLY	+0.0041	[0.0019, 0.0066]	Archival
FRESH switching	32 triplets	A_T	+0.0285	[0.0165, 0.0406]	Archival
FRESH switching	32 triplets	A_W	+0.0186	[0.0072, 0.0301]	Archival
VNF factorial	32 triplets	I_T	+0.0008	[−0.0009, 0.0030]	Archival
VNF factorial	32 triplets	I_W	+0.0010	[−0.0001, 0.0022]	Archival

3.6 Evidence levels

Each number in this paper has one of three evidence levels. Recomputed values were derived again for this paper from saved arrays: logits, probe probabilities, directions, the fitted map, and legal sets. Every recomputed bootstrap interval matches the original run to within 3×10^{-8}. That is agreement at the original seeds, not a bound on Monte Carlo error. A different seed can move the fourth digit of a 2000-replicate interval. Recorded values, such as the map's reconstruction R^2, are read from files written by the original runs and cannot be recomputed without the model checkpoint. Archival values come from verbatim console output of the original runs because the underlying arrays were not found. These are the feature-ablation, switching, and factorial results. Table 1 lists the evidence level of each headline result.

4 Results

4.1 Nullspace edits move the output while the probe readout stays fixed

The first diagnostic (FIBER) asked whether the probe's nullspace is nonempty and whether edits inside it can change next-move probabilities. The nullspace has dimension 388 of 512. That number describes this contrast matrix; it does not mean that 388 dimensions are free of board information under every decoder. On 32 matched validation pairs, adding $P_N(h_D - h_E)$ to the Editor A reference activation left the probe tensor unchanged to within 7.7×10^{-15}. It nevertheless moved the next-move distribution by mean JS 0.380, and by at least 0.185 on every pair. All 32 pairs therefore met the diagnostic's criterion of an unchanged probe readout with a non-negligible output change, and the study's letter is `PROBE_LOGIT_INSUFFICIENCY_SUPPORTED`. The 24 category-A pairs in this diagnostic share only two target games, and one target game is used in 25 of the 32 pairs.

The same archive records what happens when the kernel step is added to a natural activation rather than to h_E. These rows are scored around the natural donor activation. A donor-kernel step of L2 1 changed Y by a mean absolute 0.000316, and a same-length random kernel step by 0.000173. At L2 4 the donor-kernel mean absolute change was 0.00151 and the mean JS from the donor activation was 4.76×10^{-5}. At h_E, a correct-kernel step of mean L2 5.298 reaches JS 0.01. Kernel steps of this size therefore move the output far more at the Editor A reference than at the natural donor. The

logits for these anchor rows are not in the archive, so the means are averages of the saved per-row scores. This is not a test at the natural recipient h_R. SPEC, the matched-divergence study, and NST still have no kernel edit that starts from h_R.

On the FIBER pairs themselves, the saved factorial cells separate the probe-visible and kernel parts of $h_D - h_E$, both added to h_E. Mean Y was -0.689 at h_E, $+0.251$ after the probe-visible part, $+0.404$ after the kernel part, and $+0.758$ after both. The interaction, both components together minus the probe-visible cell minus the kernel cell plus the Editor A cell, was -0.586: the two parts both raise Y, and together they do less than the sum of the separate changes. These means come from the saved per-cell scores. They are not the VNF interaction, and they do not change VNF's letter. FIBER shows that the kernel is causally active at h_E. It does not show that the kernel carries target information rather than generic disruption, which is what the following studies test.

4.2 Correct-target nullspace directions beat L2-matched controls

SPEC applied, on 48 unused validation pairs, the correct-target nullspace direction, three wrong-target directions, and three random directions, all rescaled to the same L2 norm (mean 18.33) and added to h_E. The probe tensor stayed within 2.2×10^{-14} of its value at h_E. On the same pairs, the probe-visible part of $h_D - h_E$ had mean Y $+0.363$ and the full donor activation $+0.725$, against $+0.144$ for the kernel part. The kernel component is not the larger of the two. The mean correct-minus-wrong contrast was $+0.3916$ (interval $[0.3175, 0.4758]$, 46 of 48 pairs positive). The correct-minus-random contrast was $+0.7086$ ($[0.6021, 0.8170]$, 48 of 48). Mean Y was $+0.144$ under the correct direction, -0.248 under wrong directions, and -0.565 under random directions. The Editor A reference scored -0.639 and the natural recipient -0.707. The letter is `TARGET_SPECIFIC_NULL_EFFECT_SUPPORTED`.

Equal L2 did not equalize how much the edits moved the output. Measured from the Editor A reference, mean JS was 0.279 for correct directions, 0.223 for wrong directions, and 0.038 for random directions. A random nullspace vector of the same length barely moved the output, so the correct-minus-random contrast partly compares an edit that changes behavior with one that hardly does. The matched-divergence study addresses that problem.

4.3 Target identity still matters when output divergence is matched

The matched-divergence study scales each of the seven SPEC directions independently so that its output JS from the Editor A reference activation is 0.01. The first committed protocol for this question matched wrong and random directions to the correct direction's own divergence from h_E, and it required a mean contrast of at least 0.05. That run was inconclusive because the matching was incomplete. Its letter remains `TARGET_SPECIFIC_AT_EQUAL_JS_INCONCLUSIVE`, and we do not use its numbers. The fixed-0.01 protocol, including its floor of 0.02, was written after that outcome and on the same 48 pairs. The result below is that follow-up. It meets the later floor. It does not replace the inconclusive letter, and it is not a test of the earlier floor, which applied at a different divergence. Secondary targets of 0.005 and 0.02 are named in the fixed-0.01 protocol and were not run.

The follow-up calibrated all 336 directions on the same 48 pairs, and every pair completed. The largest deviation from the JS target was 1.23×10^{-6}. The largest change in any probe-probability entry relative to the Editor A reference was 1.19×10^{-14}. Matching divergence forced very different activation steps: mean L2 was 5.298 for correct directions, 6.368 for wrong directions, and 12.265

for random directions. Figure 2 shows, pair by pair, that the probe readout stayed at roundoff while the output distribution moved by a similar amount in every category. Mean TV distance from the Editor A reference was 0.077 for correct, 0.080 for wrong, and 0.080 for random directions.

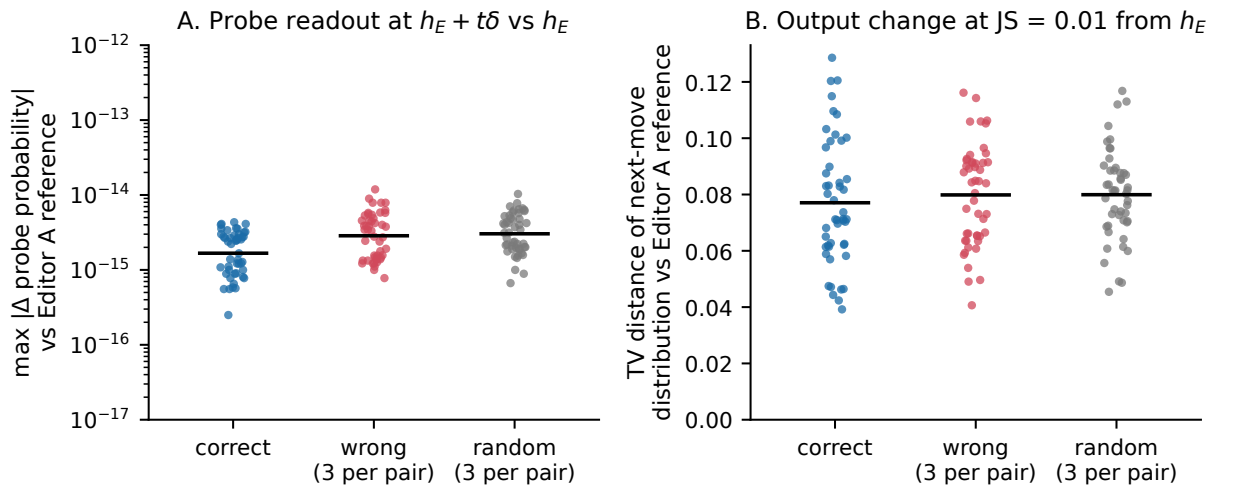


Figure 2. Probe readout and output change in the matched-divergence study, relative to the Editor A reference activation. Each point is one of 48 recipient–target pairs. For wrong and random directions, the value for a pair aggregates its three directions: the maximum in panel A and the mean in panel B. (A) Largest absolute change in any of the 64×3 probe probabilities between the patched activation $h_E + t\,\delta$ and the Editor A reference activation h_E, on a logarithmic axis. All values are below 1.2×10^{-14}. (B) Total variation distance between the next-move distributions at $h_E + t\,\delta$ and at h_E. Every direction was scaled so that its Jensen–Shannon divergence from the Editor A reference is 0.01. Horizontal bars are the geometric mean over pairs in panel A and the arithmetic mean in panel B.

Under this calibration, the mean correct-minus-wrong contrast was +0.0266 (interval $[0.0179, 0.0357]$, 40 of 48 pairs positive), and the correct-minus-random contrast was +0.0230 ($[0.0103, 0.0360]$, 30 of 48). Relative to wrong directions, the correct direction raised became-legal mass ($\Delta G = +0.0133$) and lowered became-illegal mass ($\Delta Q = -0.0133$). Mean Y was −0.592 for correct, −0.619 for wrong, and −0.615 for random directions. The letter is `FIXED_JS_TARGET_SPECIFICITY_SUPPORTED`. Figure 3 shows the pair-level contrasts and subgroups.

Three qualifications apply. First, this study reuses the SPEC pairs and is not independent confirmation. Second, the effect is a shift of 0.02–0.03 in probability mass, not recovery of the target's move distribution. Third, the advantage over random directions is not uniform across subgroups. Its interval lies above zero in category B (+0.0477, $[0.0238, 0.0722]$) and in pairs where Editor A decoded the exact target board (+0.0395, $[0.0199, 0.0584]$). Its interval includes zero in category A (+0.0106, $[-0.0033, 0.0246]$) and in pairs where Editor A did not decode the exact board (+0.0090, $[-0.0077, 0.0264]$). The advantage over wrong targets has an interval above zero in all four subgroups. The exact-board split was not in the matched-divergence preregistration. Every category-B pair is also an exact-decode pair, in this study and in NST, so those two slices are not separate evidence. The split also tracks earlier positions in NST. It is descriptive. The pair-level correct-minus-wrong contrasts here and in SPEC correlate 0.155, so reusing the pairs did not copy the SPEC result forward.

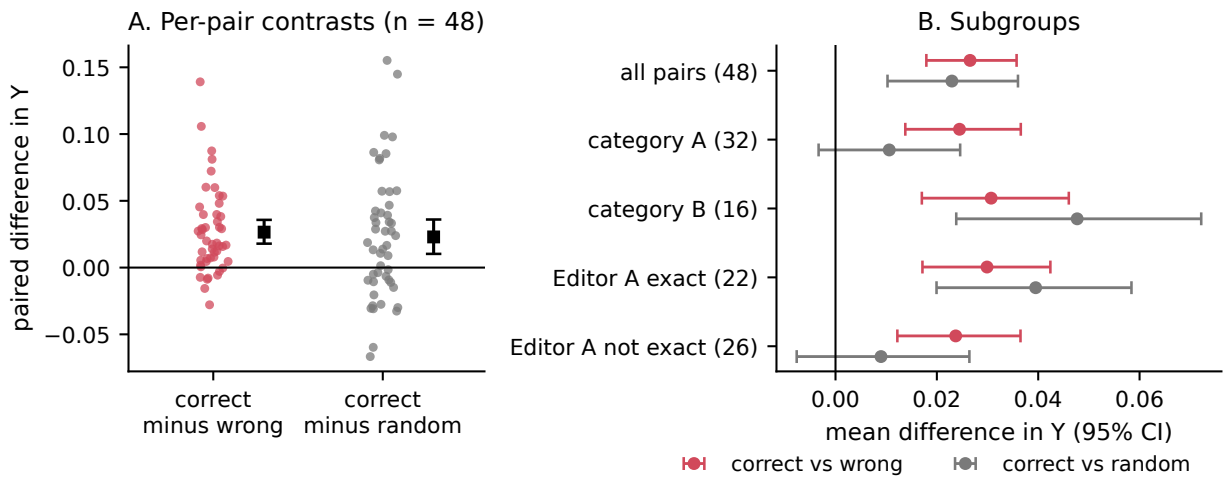


Figure 3. Paired response contrasts in the matched-divergence study. (A) Correct-minus-wrong and correct-minus-random differences in Y for each of the 48 pairs, with the mean and its clustered 95% bootstrap interval. The wrong and random responses of a pair are averaged over its three directions before the difference is taken. (B) The same means and intervals for category A (32 pairs), category B (16 pairs), and pairs on which Editor A did or did not decode the exact target board (22 and 26 pairs). Every direction was calibrated to JS 0.01 from the Editor A reference activation.

4.4 A train-only map transfers part of the effect without a target activation

Donor directions show that some kernel direction associated with the target's natural activation is useful. They do not show that such a direction can be built from the target board alone. NST fits the ridge map described above on training activations, freezes it, and evaluates it on 48 new validation pairs. At evaluation the learned direction is computed from the target's symbolic features and the Editor A reference activation. We checked that construction directly from the saved arrays. The difference between the learned and wrong-target directions equals $P_N(\hat{z}(b_t) - \hat{z}(b_w))$ computed from the saved map and the two boards' features, with relative error 3.5×10^{-15}. The difference between the learned and stage-mean directions equals $P_N\hat{z}(b_t)$ minus the saved stage mean, with relative error 3.7×10^{-15}.

The recorded residual R^2 of the map was 0.342 on training games and 0.305 on held-out validation games; reconstruction is not the causal gate. At matched divergence, the learned direction exceeded the wrong-target map direction by +0.0201 (interval $[0.0108, 0.0298]$, 39 of 48 pairs). It exceeded the random direction by +0.0314 ($[0.0185, 0.0453]$, 35 of 48), the stage-mean direction by +0.0220 ($[0.0158, 0.0293]$, 42 of 48), and the Editor A reference itself by +0.0537 ($[0.0452, 0.0618]$, 47 of 48). A random kernel direction at the same divergence already exceeds the Editor A reference by +0.0223, on 34 of 48 pairs, so part of the learned-minus-Editor-A contrast is generic disruption. Relative to Editor A, the learned edit raised became-legal mass by 0.0195 and lowered became-illegal mass by 0.0342. The absolute score stayed negative. Mean Y was −0.582 for the learned edit, −0.636 at the Editor A reference, and −0.701 at the natural recipient, and Y was positive for the learned edit in only 1 of 48 pairs. The letter is `DONOR_FREE_NULL_STATE_TRANSFER_SUPPORTED`.

"Donor-free" describes evaluation only; the map was trained on natural nullspace residuals. The learned and oracle directions behaved similarly on average (mean Y −0.582 and −0.583) but were not the same vectors. Their mean cosine similarity was 0.792. The learned direction was closer to the stage-mean direction (mean cosine 0.850) than to the oracle, and its mean cosine with the

wrong-target map direction was 0.757. Across pairs, the learned and oracle scores were highly correlated (Pearson $r = 0.998$). That correlation is not evidence of a shared mechanism. The learned score was almost as correlated with the wrong-target score ($r = 0.991$) and with the Editor A reference score ($r = 0.994$), so the correlation mostly reflects pair-to-pair differences in baseline Y. After subtracting the Editor A score, the learned score still correlates 0.846 with the oracle score and 0.452 with the wrong-target score. A gradient baseline reached the JS tolerance on only 7 of 48 pairs and is not compared. Figure 4 summarizes these results.

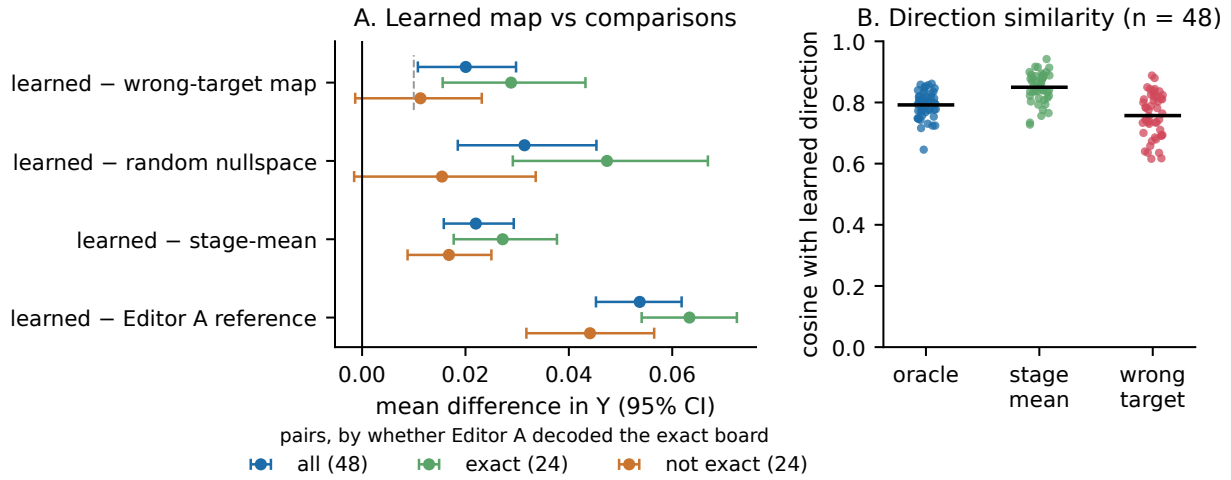


Figure 4. Donor-free transfer on 48 new validation pairs. (A) Mean difference in Y between the learned map direction and each comparison direction, with clustered 95% bootstrap intervals, for all pairs and for pairs on which Editor A did or did not decode the exact target board (24 each). All directions were calibrated to JS 0.01 from the Editor A reference activation. The dashed line marks the preregistered 0.01 floor for the learned-minus-wrong contrast. (B) Per-pair cosine similarity between the learned direction and the oracle, stage-mean, and wrong-target directions. The learned direction is closer to the stage-mean direction than to the oracle.

The transfer was weaker where Editor A failed to decode the exact target board. On those 24 pairs, the advantage over wrong targets was +0.0113 ($[-0.0013, 0.0232]$) and over random directions +0.0155 ($[-0.0015, 0.0336]$), both with intervals that include zero. On the 24 exact-board pairs the corresponding values were +0.0288 and +0.0474. We treat this split as a limitation of transfer, not as a separate finding. The 48 pairs are unused relative to SPEC but are still validation games, not confirmation.

4.5 A stage-only map accounts for part of the mean gain over Editor A

The feature-ablation study (SEM) reused the NST pairs and the frozen principal-component basis. It refitted the ridge map with feature subsets: LEGAL_ONLY drops the occupancy one-hots, BOARD_ONLY drops the legal mask, and STAGE_ONLY keeps only the player and occupancy. Because it reuses NST outcomes, it is exploratory. Its full-feature arm is the NST learned map. The console value of SEM's FULL-minus-Editor A contrast, 0.053713, agrees with our recomputation from the NST arrays to 7×10^{-9}.

The archival values were as follows. FULL minus LEGAL_ONLY was +0.0041 ($[0.0019, 0.0066]$) and FULL minus BOARD_ONLY was +0.0056 ($[0.0021, 0.0090]$); both are below the preregis-

tered +0.01 floor for an incremental feature advantage. BOARD_ONLY minus STAGE_ONLY was +0.0180 and LEGAL_ONLY minus STAGE_ONLY was +0.0195. The letter is `LEGAL_MASK_SUFFICES_IN_TESTED_FAMILY`. On these pairs the console's forward switching contrast is the learned-minus-wrong contrast already reported. Its reverse contrast is $A_W = +0.0058$ (interval $[-0.0003, 0.0124]$). That interval includes zero. The letter does not mean that the legal mask is uniquely necessary. In Othello, occupancy determines legality, so a board one-hot and a legal mask are largely redundant.

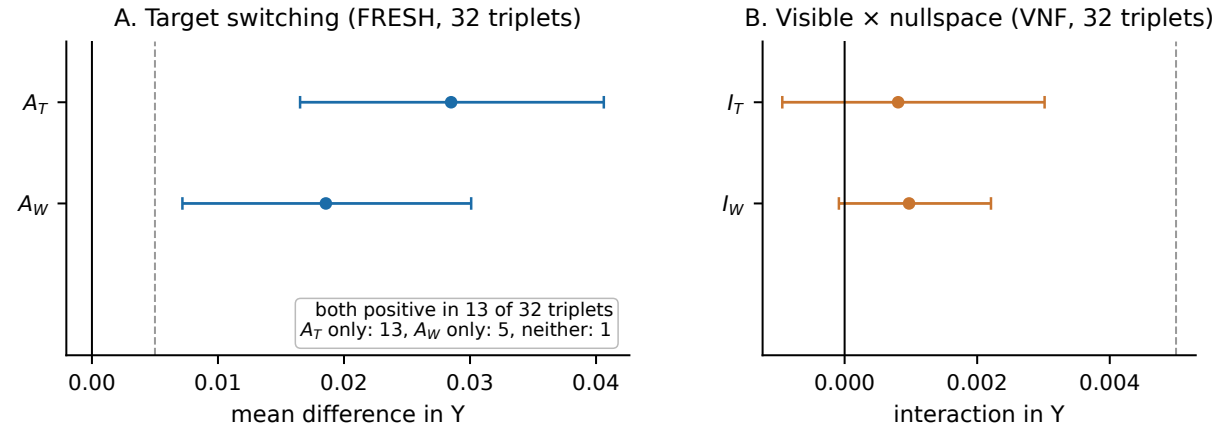
The same values give a mean-level identity. FULL minus STAGE_ONLY is $0.0041 + 0.0195 = 0.0236$ through the legal-only arm and $0.0056 + 0.0180 = 0.0236$ through the board-only arm. Combined with the recomputed FULL-minus-Editor A value of 0.0537, STAGE_ONLY minus Editor A is 0.0301. STAGE_ONLY therefore accounts for 56.04% of FULL's mean improvement over Editor A. That ratio uses Editor A as the baseline. A random kernel direction already accounts for +0.0223 of the learned map's +0.0537 gain over Editor A, and the stage-only gain over Editor A is 0.0301, so most of the stage-only gain is shared with an unstructured kernel edit. The learned-minus-random contrast of +0.0314 is the comparison that removes that disruption. The archival subgroup shares were 58.5% in category A, 52.6% in category B, 56.8% on exact-board pairs, and 55.0% on nonexact pairs. This is a ratio of means under a shared calibration, not a decomposition into independent mechanisms. STAGE_ONLY is conditioned on the intended target's player and occupancy, so it is not a target-blind prior. Its low reconstruction R^2 describes how well it reconstructs nullspace residuals, not its share of the mean response, and does not justify discounting its behavioral contribution. Taken together with the NST results, the donor-free edit is better described as a target-conditioned nudge than as a 64-component code for the board. Much of its average gain over Editor A is shared with a random kernel direction, so it is not by itself evidence of a stage-specific correction, and its additional content over legal-only or board-only maps is below the study's own floor.

4.6 Reliable target switching and a visible–nullspace interaction were not supported

If the learned nullspace edit carried a transferable target state, requesting board T rather than board W on the same recipient should move Y toward whichever target was requested, often enough to pass a joint gate. For a triplet (R, T, W) let A_T be the change in Y scored on T's legal sets when the edit requests T rather than W. Let A_W be the corresponding change scored on W's sets when the edit requests W rather than T. The remaining validation games contained 2 category-B pairs but no category-B triplets, so the study was first recorded as `SYMMETRIC_TARGET_CENSUS_INFEASIBLE`. A new cohort generated from seed 9182 supplied 16 A and 16 B triplets. All values in this section are archival.

With the edit applied to the natural recipient activation and calibrated to JS 0.01, A_T was +0.0285 ($[0.0165, 0.0406]$, positive in 81% of triplets). A_W was +0.0186 ($[0.0072, 0.0301]$, positive in 56%). Both were positive together in 13 of 32 triplets (40.6%), below the preregistered 60% gate. Every triplet completed. The letter is `SYMMETRIC_TARGET_SWITCHING_NOT_SUPPORTED`. The same console record marks a second gate, anchor dependence, as failed. The Editor-A-anchored switching values are not in the excerpt, so they are not reported here. Positive average discrimination and reliable simultaneous switching are different claims; the gate tests the second, and it failed. Editor A's own visible contrasts on the same triplets were small and positive ($D_T = +0.0158$, $D_W = +0.0151$), with lower interval bounds of 0.0002 and 0.0018. They depend on how well Editor A decoded each board and are not a test of compatibility between targets.

The factorial study asked a cleaner interaction question on another new cohort (seed 10182). An independently fitted rowspace ridge map and the frozen nullspace map were combined with fixed coefficients ($\rho = \nu = 3.5$), so that all four target-by-component cells had the same total L2 and no divergence calibration was involved. The interaction estimands were $I_T = d_T(T) - d_T(W)$ and $I_W = d_W(W) - d_W(T)$. The archival estimates were $I_T = +0.0008$ ($[-0.0009, 0.0030]$, positive in 17 of 32 triplets) and $I_W = +0.0010$ ($[-0.0001, 0.0022]$, 19 of 32). Both fall below the preregistered $+0.005$ floor, and both intervals include zero. The letter is `TARGET_SPECIFIC_VISIBLE_NULL_INTERACTION_NOT_SUPPORTED`. The rowspace map was not empty: its held-out reconstruction R^2 for $P_R h$ was 0.9405. Reconstructing the probe-visible state accurately did not produce the hypothesized interaction in Y. Figure 5 summarizes both studies.



Archival values from the original console output; dashed line: preregistered 0.005 floor.

Figure 5. Completeness tests that failed (archival values from the original console output). (A) Target-switching contrasts A_T and A_W on 32 fresh triplets, with 95% bootstrap intervals and the preregistered 0.005 floor (dashed). The box in panel A gives how many triplets had both contrasts positive, only one positive, or neither. Both-positive switching occurred in 13 of 32 triplets, below the 60% gate. (B) Visible–nullspace interaction estimates I_T and I_W on a second fresh cohort of 32 triplets, with intervals and the 0.005 floor.

These failures belong in the main argument. They are why we do not claim to have recovered a transferable decision state.

4.7 The original board-edit validation remains negative

Before the kernel studies, the 08R2 validation asked whether Editor A repairs legal-move behavior on frozen validation pairs. Its letter is `08R2_VALIDATION_COMPLETE — NEGATIVE`, and its pair-level tables are not part of this package. The result still bounds the claims here. Optimizing a residual toward a decoded board with that editor did not produce the behavioral repair the protocol required. Nullspace edits are a different intervention family and should not be read as having solved that failure. They produce small, target-aligned contrasts in Y under matched controls, while absolute scores remain negative and switching fails its joint gate.

5 Discussion

For this frozen linear board probe, at the Editor A reference activation, the residual stream is not a single occupancy code plus unused orthogonal content. That sentence is about h_E. At the natural donor activation, kernel steps of similar length barely moved the output. Starting from an activation whose probe readout has already been moved toward a target board, directions that leave the probe's 64×3 probabilities unchanged still change legal-move probability. They change it more for the intended target than for wrong targets, whether residual L2 or output divergence is matched. That is a statement about one probe, one site, and one model. It agrees with the warning of Elazar et al. that readout is not a causal summary, and it agrees with Li et al. and Nanda et al. that board features are causally relevant, because those features are this probe's rowspace. The kernel results say that the rowspace of this probe is not the whole causal story.

The donor-free result adds something that patching from h_D cannot. A linear map from simulator features into the kernel predicts a direction that keeps a target-aligned advantage over wrong-target and random controls. The map's inputs include the true legal mask and the true occupancy. So the result does not mean that the network taught us the rules. It means that, given the rules and the board, a map trained only on training games builds a probe-preserving edit that is on average better than the same map's edit for a wrong board. Stage-only features already deliver more than half of the gain over Editor A, and much of that half is shared with a random kernel direction, so the sentence overstates the amount of board-specific geometry in the map.

Several tests that a sufficient decision state would pass were failed. Absolute Y stays negative. Joint switching fails its 60% gate. The visible–nullspace interaction is indistinguishable from zero at the preregistered floor. We have not shown that $h_E + t\,\delta$ lies on the natural activation manifold, only that this probe cannot distinguish it from h_E, and Editor A itself may produce unnatural activations. Matching divergence does not equalize L2, so every matching is relative to one scalar. The matched-divergence study reuses the SPEC pairs. In the FRESH cohort of 400 games, 24 board-and-player states also occur in earlier splits. Whether any of those states appear among the 32 selected triplets cannot be checked without the triplet arrays. Game identifiers in the selected set are disjoint from earlier studies by construction.

Nothing here is evidence about large language models, factual recall, or safety properties of deployed systems. Othello is deterministic and fully observed, and every score uses simulator legal sets.

6 Limitations

The study uses one architecture, one checkpoint, one residual site, and one frozen probe. Synthetic legal self-play is neither human Othello nor natural language. Target labels and legal masks come from a simulator, so methods that receive those masks are not inferring legality.

Probe invariance is defined relative to this probe's probability tensor at the Editor A reference activation. Another probe, a nonlinear decoder, or a readout at a later layer might detect the edit. Because every edit in SPEC, the matched-divergence study, and NST starts from h_E rather than h_R, the effects are conditional on Editor A's change to the decoded board. Those three studies were not rerun from h_R. The FIBER diagnostic did perturb the natural donor activation, and kernel steps there barely moved the output, as reported above. Nothing here measures kernel effects at the natural recipient.

Nullspace addition can leave the natural activation manifold. We report no density model or

cycle-consistency check that would rule that out.

After divergence matching, effects on Y are a few hundredths of probability mass, and mean learned Y is about -0.58. Intervals that exclude zero describe a small, consistent contrast, not a large behavioral repair.

Calibration choices change the scientific object. SPEC's correct-minus-random contrast is partly a comparison against weak random disruption. Divergence matching shrinks the contrast and enlarges random L2. The factorial study matches L2 and finds no interaction. These are different interventions, not contradictory measurements of one quantity.

Evaluation samples are reused. SPEC and the matched-divergence study share 48 pairs, and NST and SEM share another 48. Only FRESH and VNF use later independent cohorts, and those are the studies whose completeness tests failed. Sealed confirmation data were not used, and no independent external replication has been performed.

The ablation, switching, and factorial results rest on console output from the original runs. Their pair- and triplet-level arrays were not found, so their means and intervals could not be recomputed for this paper. We report them with that label rather than omit them, because two of them are the negative results that bound our claims.

Wrong-target selection uses simulator features, not model responses, as intended. Pairs require nonempty legal-set differences, so they are not a uniform sample of positions. No pairs were excluded after seeing Y.

7 Conclusion

On Othello-GPT, a frozen linear board probe can be held fixed, in the strong sense of its full probability tensor at the Editor A reference activation, while residual edits in the kernel of its contrasts still change legal-move predictions. Some of those changes are target-aligned against wrong-target and random controls, including when output divergence is matched. A map trained only on training games produces part of the effect without a target activation at evaluation. The same evidence shows that the map is not a transferable board state. Absolute response scores stay negative, more than half of the gain over Editor A is stage-conditioned, joint switching fails its gate, and a visible–nullspace interaction is not supported. The justified conclusion is the limited one, and it is limited to edits at the Editor A reference: this probe is not a sufficient causal description of the residual information that affects legal-move outputs there. At a natural donor activation, kernel steps of similar length barely moved the output. The results do not show that the model has a second world model.

8 Reproducibility and data availability

The accompanying package contains the saved logits, probe probabilities, intervention directions, fitted map, legal sets, and pair manifests for the FIBER, SPEC, matched-divergence, and NST studies. It also contains scripts that recompute every value labeled recomputed, regenerate all tables and figures, and check every number in this manuscript against its source. The model checkpoint and probe weights are not redistributed, so replaying interventions requires obtaining them separately. Results from the feature-ablation, switching, and factorial studies are included as verbatim console excerpts with hashes, and their arrays remain to be located. Protocols and

configuration files for every study are included.

This work used AI assistance for engineering, auditing, and drafting. An AI system is not an author. Authors, affiliations, and a disclosure statement will be written by the people responsible for the project.

Official result letters, unchanged: `08R2_VALIDATION_COMPLETE — NEGATIVE`; `TARGET_SPECIFIC_NULL_EFFECT_SUPPORTED`; `TARGET_SPECIFIC_AT_EQUAL_JS_INCONCLUSIVE`; `FIXED_JS_TARGET_SPECIFICITY_SUPPORTED`; `DONOR_FREE_NULL_STATE_TRANSFER_SUPPORTED`; `LEGAL_MASK_SUFFICES_IN_TESTED_FAMILY`; `SYMMETRIC_TARGET_CENSUS_INFEASIBLE`; `SYMMETRIC_TARGET_SWITCHING_NOT_SUPPORTED`; `TARGET_SPECIFIC_VISIBLE_NULL_INTERACTION_NOT_SUPPORTED`; `OA = C / PARTIAL`; `SECTION_5_3 = PARTIAL`; `OBSTRUCTION_READY = NO`.

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A Definitions and protocol details

Token mapping. Square $s \in \{0, \ldots, 63\}$ in row-major order maps to token $s + 1 - c(s)$, where $c(s)$ is the number of center squares (27, 28, 35, 36) below s. This mapping reproduces every saved legal-token list in the NST archive.

Editor A. Editor A minimizes a board-decoding loss for the target's player-relative labels under the frozen probe, starting from h_R, with an L2 budget of 16, 40 steps, learning rate 0.4, and norm penalty 0.01. Its output is the Editor A reference activation h_E. Editor A decoded the exact target board on 22 of the 48 SPEC pairs and on 24 of the 48 NST pairs. Its logits are bitwise identical in the SPEC and matched-divergence archives, so both studies share the same reference.

Wrong-target and random directions. Each SPEC pair has three wrong-target donors with the same player and with occupancy, legal-set size, and prefix length matched within a slack of 4, selected by frozen seeds before any Y was computed. Random directions are $P_N g$ for seeded Gaussian g. In SPEC, all six controls are rescaled to the correct direction's L2 norm, to relative error 8×10^{-16}. In the matched-divergence study, each is scaled independently to JS 0.01 from h_E.

Divergence search. The frozen grid is 0.003125, 0.00625, 0.0125, 0.025, 0.05, 0.1, 0.2, 0.4, 0.8, 1.0, 1.5, 2.0, 4.0, 8.0. The first grid point with JS at least 0.01 is bisected 16 times against its predecessor. A direction that cannot reach 0.01 fails and remains in the attempt record. No direction failed in the matched-divergence or NST studies, except the gradient baseline, which calibrated on 7 of 48 pairs.

Preregistered gates. SPEC required a mean correct-minus-wrong contrast of at least 0.05 with interval above zero, $\Delta G > 0$ and $\Delta Q < 0$ relative to wrong directions, and a correct-minus-random contrast with interval above zero. The matched-divergence study required 44 of 48 complete pairs, a mean correct-minus-wrong contrast of at least 0.02, both intervals above zero, $\Delta G > 0$, and $\Delta Q < 0$. NST required a learned-minus-wrong contrast of at least 0.01, intervals above zero against wrong, random, stage-mean, and Editor A, and the intended ΔG and ΔQ signs relative to Editor A. SEM required an incremental advantage of at least 0.01 with interval above zero. FRESH required at least 29 of 32 complete triplets, A_T and A_W of at least 0.005 with intervals above zero, a positive mean of $C = A_T + A_W$, and both contrasts positive on at least 60% of triplets. The archival mean of C was +0.0471 ($[0.0316, 0.0629]$); the protocol states that a positive C alone is not bidirectional switching. VNF required at least 29 of 32 complete triplets, I_T and I_W of at least 0.005 with intervals above zero, and random and label-shuffled controls that do not reproduce that pattern. Because the interaction floor was not met, the control condition does not affect the outcome.

Bootstrap. Clusters are recipient games, taken in order of first appearance, resampled with replacement using NumPy's default generator. The statistic is the mean of cluster means, and intervals are the 2.5th and 97.5th percentiles. Seeds are 82 plus a metric offset; NST subgroups add 10, 20, 30, or 40. FRESH and VNF used seeds 9482 and 10482 with 5000 replicates.

B Additional recomputed results

Table 2 summarizes the 336 matched-divergence interventions by category. Every saved Y value matched our recomputation to 1.3×10^{-7}, every saved JS value to 4.4×10^{-9}, and every saved L2 to relative error 4.4×10^{-16}. The differences come from float32 storage of probabilities in the original runs.

For NST, the subgroup contrasts by category were as follows. In category A the learned-minus-

Table 2. The 336 matched-divergence interventions by direction category, recomputed from saved arrays. TV distance and probe change are measured relative to the Editor A reference activation. TV is averaged over interventions; the probe change is the largest absolute change in any of the 64×3 probe probabilities over all interventions in the category.

Category	Mean Y	Mean L2	Mean TV from Editor A	Max probe change
correct	−0.5925	5.298	0.0771	4.3e−15
wrong	−0.6190	6.368	0.0799	1.2e−14
random	−0.6154	12.265	0.0799	1.0e−14

wrong contrast was +0.0181 ([0.0074, 0.0294]); in category B it was +0.0239 ([0.0075, 0.0416]). Against random directions the values were +0.0188 ([0.0041, 0.0338]) in A and +0.0566 ([0.0331, 0.0798]) in B. The natural recipient activation h_R and the learned patched activation differ by mean JS 0.045. The Editor A reference and h_R differ by mean JS 0.030.

C Provenance

The saved arrays come from four result archives produced by the original Colab runs. Their SHA-256 hashes and the hash of every member file are listed in the package's provenance manifest. Protocol and configuration files were copied from the project repository with their git blob identifiers. The archival values come from eleven verbatim console excerpts, each hashed. A search of the local file system, including all downloaded archives, cloud-synchronized folders, and the repository artifact tree, found no result arrays for the feature-ablation, switching, or factorial studies. No Google Drive mount was available on the machine used to prepare this package.